SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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CSA37- SOFTWARE TESTING

EXP:6

Name: Balaguru S

Register Number: 192312070

Requirement Analysis and Positive Test Scenarios and Test Cases for a Food Delivery Application

Aim

To study the requirements of a Food Delivery Application, and to prepare positive test scenarios and test cases that confirm the application works correctly for valid inputs and normal user actions.

Problem Statement

A food delivery application lets customers choose a restaurant, order food, pay online and follow the delivery on a map. It depends on location services, payments and real-time updates, so a defect can mean cold or wrong food, wrong charges or failed orders.

Testing is needed to check that the application:

- Customers can log in, order food and track the delivery correctly.
- Invalid inputs and failures are handled safely.
- Location, cart, payment and cancellation rules follow the requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	The user can register and log in with a mobile number and a one-time password (OTP).
FR2	The app detects the user's location or accepts a manual delivery address inside the service area.
FR3	The user can browse restaurants and filter by cuisine, rating, veg only and delivery time.
FR4	The restaurant menu shows item names, prices, veg or non-veg marks and customization options.
FR5	The user can add items to the cart, change quantities and customise items. Unavailable items cannot be added.
FR6	The user can apply a valid coupon. A coupon works only above its minimum order value.
FR7	Payment is possible by UPI, card, wallet or cash on delivery.
FR8	After payment, the app confirms the order and shows an estimated delivery time.
FR9	The user can track the order status and the delivery partner's live location.
FR10	An order can be cancelled only before the restaurant starts preparing it.
FR11	After delivery, the user can rate the restaurant and the delivery partner.
FR12	A cancelled or failed paid order is refunded to the original payment method.

ID	Requirement
FR13	The user can view past orders and reorder from the history.

2 Software and environment requirements

Item	Details
Application	Android and iOS mobile app, with a web version
Back end and APIs	Java/Node.js services, Google Maps API, SMS gateway for OTP
Database	MySQL / MongoDB (users, restaurants, menus, orders)
Operating system	Android 12+ and iOS 16+ (test devices or emulators)
Testing tools	Test case document (Excel), Appium, Postman, bug tracker (Jira)
Test data	Test users, restaurants with menus, coupon codes, delivery addresses inside and outside the service area, sandbox payment details

Positive Test Scenarios

A positive test scenario checks that a feature works as expected when the user gives valid input and follows the normal flow.

Scenario ID	Module	Test Scenario	Requirement
FD-PS01	Login	Log in with a valid mobile number and OTP	FR1
FD-PS02	Location	Set the delivery address using GPS	FR2
FD-PS03	Browse	Filter restaurants by veg and rating	FR3
FD-PS04	Menu	View a restaurant menu	FR4
FD-PS05	Cart	Add a customised item to the cart	FR5
FD-PS06	Cart	Change the quantity and remove an item	FR5
FD-PS07	Offers	Apply a valid coupon	FR6
FD-PS08	Payment	Pay using UPI	FR7
FD-PS09	Payment	Place an order with cash on delivery	FR7, FR8
FD-PS10	Tracking	Track the order live	FR9
FD-PS11	Cancellation	Cancel an order before preparation starts	FR10, FR12
FD-PS12	Rating	Rate the restaurant and delivery partner	FR11
FD-PS13	History	Reorder from the order history	FR13

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
FD-PT01	FD-PS01	1. Open the app 2. Enter the mobile number and tap Get OTP 3. Enter the OTP received 4. Tap Verify	9876543210; OTP received by SMS	User is logged in and the home screen is displayed.
FD-PT02	FD-PS02	1. Allow location permission 2. Confirm the detected address	Location inside the service area	Address is detected and nearby restaurants are listed.
FD-PT03	FD-PS03	1. Open the home screen 2. Apply Pure Veg and Rating 4.0+ filters	Veg only; rating 4.0+	Only veg restaurants rated 4.0 or above are shown.
FD-PT04	FD-PS04	1. Select a restaurant from the list	Any open restaurant	Menu shows item names, prices, veg marks and customization options.
FD-PT05	FD-PS05	1. Open a menu item 2. Choose a customization 3. Tap Add to Cart 4. Open the cart	Margherita pizza; extra cheese	Item is added with the customization and the correct price.
FD-PT06	FD-PS06	1. Open the cart 2. Increase the quantity to 3 3. Reduce it back to 0	Quantity 3, then 0	Total is updated each time. Item is removed at quantity 0.
FD-PT07	FD-PS07	1. Add items worth ₹400 2. Tap Apply Coupon 3. Enter the code	FIRST50 (₹50 off above ₹300)	Discount of ₹50 is applied to the bill.
FD-PT08	FD-PS08	1. Tap Proceed to Pay 2. Choose UPI 3. Approve the sandbox payment	success@upi	Payment is successful and the order is placed.
FD-PT09	FD-PS09	1. Choose Cash on Delivery 2. Tap Place Order	Order value ₹450	Order is confirmed with an estimated delivery time.
FD-PT10	FD-PS10	1. Open Orders 2. Select the active order 3. View the tracking screen	Active order	Status moves from Preparing to Out for Delivery, and the partner’s location is shown on the map.
FD-PT11	FD-PS11	1. Place a prepaid order 2. Tap Cancel before the restaurant accepts 3. Confirm	Prepaid order ₹450	Order is cancelled and a refund to the original payment method is initiated.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
FD-PT12	FD-PS12	1. Open a delivered order 2. Select star ratings 3. Add a comment 4. Submit	4 stars; comment "Tasty and on time"	Rating is saved and shown in the order details.
FD-PT13	FD-PS13	1. Open Order History 2. Select a past order 3. Tap Reorder	Previous order	Items are added to the cart with the current prices.

Result / Conclusion

Result: 13 positive test scenarios and 13 test cases were prepared for the Food Delivery Application, covering 13 of the 13 functional requirements (FR1 to FR13). The test cases by module are:

Module	Test Scenarios	Test Cases
Login	1	1
Location	1	1
Browse	1	1
Menu	1	1
Cart	2	2
Offers	1	1
Payment	2	2
Tracking	1	1
Cancellation	1	1
Rating	1	1
History	1	1
Total	13	13

Conclusion: The positive test cases cover the main user flows of the Food Delivery Application, from the first action to the final result. When they are executed on a build, they confirm that the valid-input behaviour matches the specification. They should be run first in every test cycle and kept in the regression suite, with the negative test cases added to check invalid inputs.